Calibrated freight proxy sensitivity: one-year lag

direct truck

direct rail

inland barge

Rodeo rail gulf marine ocean0.5

Rodeo rail gulf marine ocean1

Rodeo rail gulf marine ocean2

Rodeo river gulf marine ocean0.5

Rodeo river gulf marine ocean1

Rodeo river gulf marine ocean2

rail with Lemont road access

rail with StRose road access

rail Lemont barge StRose marine Rodeo

rail via Artesia pretreatment

0–25
25–50
50–75
75–100
100–125
125–150

Exploratory TWFE, 2014–2025; crusher-clustered 95% intervals. Linehaul only; excludes handling/access. Multiple testing; not causal.
Relative decay: 100 truck-equivalent km; assumed scale. 2025 dose uses 2024 crush in all known-county bins.

Soybean-share percentage points per 100 million gal/year exposure